

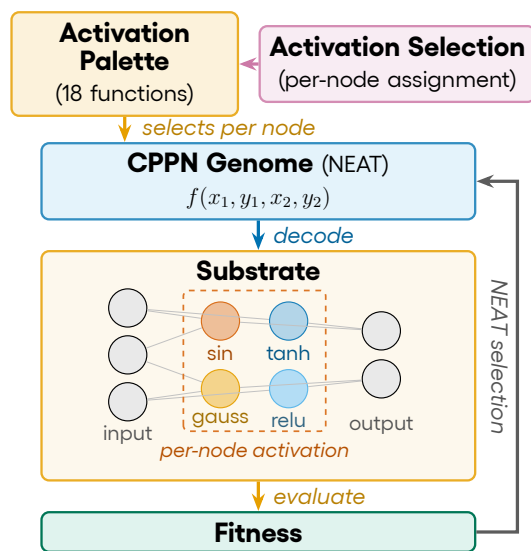
# The activation function decides what is solvable.

Know which one and the task is easy. When you don't, evolution can find it, one function per node.

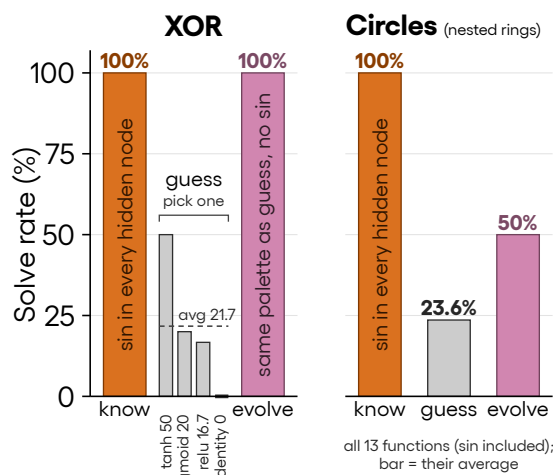
## Per-Node Activation Function Evolution in Indirectly Encoded Substrates: Solvability, Limits, and Emergent Diversity

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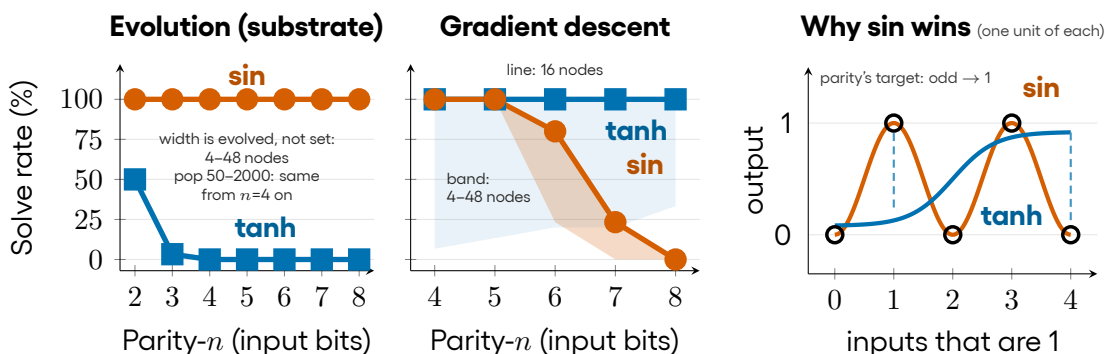
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### Know it, guess it, or evolve it?



**Know:** sin, a right answer, in every hidden node. **Guess:** one function for all nodes. **Evolve:** the CPPN picks per node. **XOR's palette:** tanh, sigmoid, relu, identity. **No sin, yet evolution matches sin anyway.** **Circles** has all 13 with sin; that freedom costs 50% vs 100%. 30 replications each.



**Why you often don't know:** the right choice inverts with the optimizer (30 replications each).  
**Why sin wins:** parity alternates; a turning curve follows, a flat one cannot (dashed: misses).

### The problem

Brains mix cell types. Networks give every node the same activation, and that choice decides what the network **can** represent: on Parity-4, 4 of 18 functions always solve and 9 never do. More turns in a function's curve, more often it solves ( $\rho=0.81$ ).

### The approach

An **extra CPPN output channel** gives every node its own function from an 18-function palette; panels use 4- and 13-function subsets. 4,500+ runs: Boolean, regression, and spatial tasks, all  $\leq 8$  inputs.

### Findings

**Know it, it's trivial:** sin solves Parity-4 in **1.7 gens**.

**Guess it, it fails:** the defaults tanh, relu, sigmoid score **0%** on Parity-4.

**Evolve it, it works:** a palette without sin still solves XOR **30/30**, at  $16 \times$  sin's gens.

### Why open-ended?

Nobody picks the functions: evolution assigns them node by node, building networks no designer would create.

### Where it stops

Evolution cannot find what the palette lacks: XOR's four flat defaults score **0%** on circles. Gradient descent needs width: **2/30** to **30/30**.

### TAKE-HOME

**When you don't know, don't pick a function. Give evolution a palette.**

